

QUIZ 2

TRANSPORT PHENOMENA I ChE 311 Fall 2025

Closed Book, Time: 15 minutes

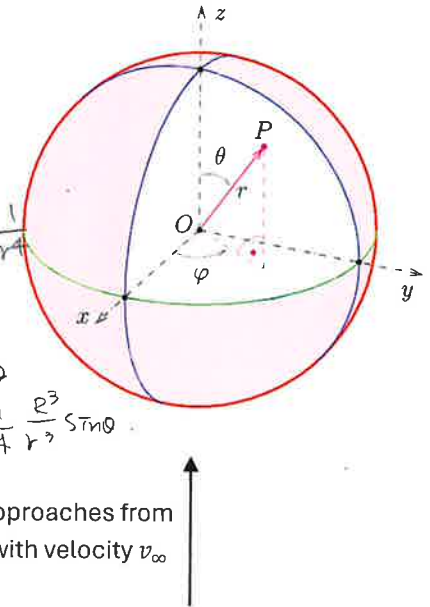
A fluid of constant density ρ and constant viscosity μ approaches a fixed sphere of radius R with a constant velocity v_∞ vertically upward in z -direction.

The velocity profile developed around the sphere is found to be:

$$v_r = v_\infty \left[1 - \frac{3}{2} \left(\frac{R}{r} \right) + \frac{1}{2} \left(\frac{R}{r} \right)^3 \right] \cos \theta$$

$$v_\theta = v_\infty \left[-1 + \frac{3}{4} \left(\frac{R}{r} \right) + \frac{1}{4} \left(\frac{R}{r} \right)^3 \right] \sin \theta$$

$$v_\phi = 0$$



Fluid approaches from below with velocity v_∞

Using Newton's law of viscosity in spherical coordinates, find the three components of the viscous stress acting on the surface of the sphere.

$$\tau_{rr} = \frac{\partial v_r}{\partial r} = v_\infty \left[\frac{3}{2} R \cdot \frac{1}{r^2} - \frac{3}{2} R^3 \frac{1}{r^4} \right] \cos \theta$$

$$\tau_{r\theta} = \frac{\partial v_\theta}{\partial r}$$

$$\tau_{r\phi} = \frac{\partial v_\phi}{\partial r}$$

$$\tau_{\theta r} = \frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r}$$

$$\tau_{\theta\theta} = \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r}$$

$$\tau_{\theta\phi} = \frac{1}{r} \frac{\partial v_\phi}{\partial \theta}$$

$$\tau_{\phi r} = \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\phi}{r}$$

$$\tau_{\phi\theta} = \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} - \frac{v_\phi}{r} \cot \theta$$

$$\tau_{\phi\phi} = \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta}{r} \cot \theta$$

Name:

55

Shiloh

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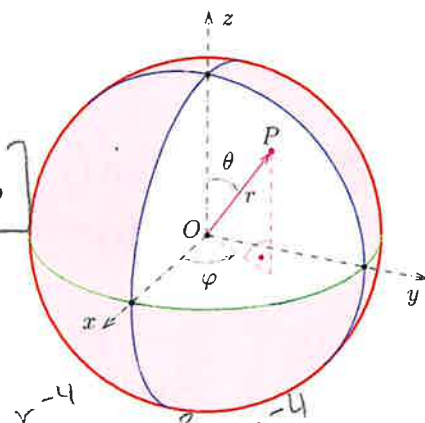
A fluid of constant density ρ and constant viscosity μ approaches a fixed sphere of radius R with a constant velocity v_∞ vertically upward in z -direction.

The velocity profile developed around the sphere is found to be:

differentiate w.r.t $r \rightarrow$

$$v_r = v_\infty \cdot \cos \theta \left[1 - \frac{3R}{2} r^{-1} + \frac{1}{2} R^3 r^{-3} \right]$$

$$v_\theta = v_\infty \left[-1 + \frac{3}{4} \left(\frac{R}{r} \right) + \frac{1}{4} \left(\frac{R}{r} \right)^3 \right] \sin \theta$$

$$v_\phi = 0$$


Fluid approaches from below with velocity v_∞

Using Newton's law of viscosity in spherical coordinates, find the three components of the viscous stress acting on the surface of the sphere.

$$\{\nabla \mathbf{v}\}_{rr} = \frac{\partial v_r}{\partial r}$$

$$\{\nabla \mathbf{v}\}_{r\theta} = \frac{\partial v_\theta}{\partial r}$$

$$\{\nabla \mathbf{v}\}_{\theta\theta} = \frac{\partial v_\theta}{\partial \theta}$$

$$\{\nabla \mathbf{v}\}_{r\phi} = \frac{1}{r} \frac{\partial v_\phi}{\partial \theta} - \frac{v_\phi}{r}$$

$$\{\nabla \mathbf{v}\}_{\theta\phi} = \frac{1}{r} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r}$$

$$\{\nabla \mathbf{v}\}_{\phi\phi} = \frac{1}{r} \frac{\partial v_\phi}{\partial \phi}$$

$$\{\nabla \mathbf{v}\}_{\phi r} = \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\phi}{r}$$

$$\{\nabla \mathbf{v}\}_{\phi\theta} = \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} - \frac{v_\phi}{r} \cot \theta$$

$$\{\nabla \mathbf{v}\}_{\phi\phi} = \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta}{r} \cot \theta$$

$$|\tau| = -\mu(|\nabla \mathbf{v}| + |\nabla \mathbf{v}|^T) + \left(\frac{2}{3}\mu - \kappa\right)(\text{div } \mathbf{v})|\delta|$$

$$\frac{\partial v_r}{\partial r} = v_\infty \cdot \cos \theta \left[0 + \frac{3R}{2} r^{-2} - \frac{3}{2} R^3 \cdot r^{-4} \right]$$

$$= v_\infty \cos \theta \left[\frac{3R}{2r^2} - \frac{3R^3}{2r^4} \right]$$

$$\tau_{rr} = -2\mu \left[v_\infty \cos \theta \left[\frac{3R}{2r^2} - \frac{3R^3}{2r^4} \right] \right]$$

$$\tau_{rr} = -2\mu v_\infty \cdot \cos \theta \left[-\frac{3R}{r^2} - \frac{3R^3}{r^4} \right]$$

$$\tau_{rr}|_{r=R} = -2\mu v_\infty \cdot \cos \theta \left[-\frac{3R}{R^2} + \frac{3R^3}{R^4} \right] = 0$$

$$-\frac{3}{R} + \frac{3}{R} = 0$$

→ Back

$$V_\theta = V_\infty \left[-1 + \frac{3}{4} \left(\frac{R}{r} \right) + \frac{1}{4} \left(\frac{R}{r} \right)^3 \right] \sin \theta$$

$$\tau_{\theta r} = -\mu \left[r \frac{\partial}{\partial r} \left(\frac{V_\theta}{r} \right) + \frac{1}{r} \frac{\partial V_r}{\partial \theta} \right]$$

$$\rightarrow \frac{V_\theta}{r} = V_\infty \left[-\frac{1}{r} + \frac{3R}{4} r^{-2} + \frac{1}{4} R r^{-4} \right] \sin \theta$$

$$\frac{V_\theta}{r} = \underbrace{V_\infty \cdot \sin \theta \left(-r^{-1} + \frac{3R}{4} r^{-2} \right)}_{\text{Differentiate first}} + \underbrace{V_\infty \cdot \sin \theta \left(\frac{1}{4} R r^{-4} \right)}_{\text{Differentiate second}} \rightarrow \text{Then will add together}$$

↳ w.r.t θ

$$\frac{\partial}{\partial r} \frac{V_\theta}{r} = V_\infty \cdot \sin \left(r^{-2} - \frac{3R}{2} r^{-3} \right)$$

$$\rightarrow \frac{\partial}{\partial \theta} \frac{V_\theta}{r} = V_\infty$$

Differentiate
Second term
w.r.t θ

Name: Lucia Hernandez

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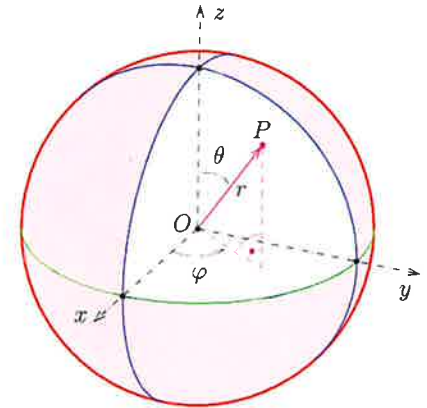
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$$v_\theta = v_\infty \left[-1 + \frac{3}{4} \left(\frac{R}{r} \right) + \frac{1}{4} \left(\frac{R}{r} \right)^3 \right] \sin \theta$$

$$v_\phi = 0$$



Fluid approaches from below with velocity v_∞

Using Newton's law of viscosity in spherical coordinates, find the three components of the viscous stress acting on the surface of the sphere.

$$\{\nabla \mathbf{v}\}_{,r} = \frac{\partial v_r}{\partial r}$$

$$\{\nabla \mathbf{v}\}_{,\theta} = \frac{\partial v_\theta}{\partial \theta}$$

$$\{\nabla \mathbf{v}\}_{,\phi} = \frac{\partial v_\phi}{\partial \phi}$$

$$\{\nabla \mathbf{v}\}_{,r\theta} = \frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r}$$

$$\{\nabla \mathbf{v}\}_{,\theta r} = \frac{1}{r} \frac{\partial v_\theta}{\partial r} + \frac{v_r}{r}$$

$$\{\nabla \mathbf{v}\}_{,\phi\phi} = \frac{1}{r} \frac{\partial v_\phi}{\partial \phi}$$

$$\{\nabla \mathbf{v}\}_{,\phi r} = \frac{1}{r \sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{v_\phi}{r}$$

$$\{\nabla \mathbf{v}\}_{,\phi\theta} = \frac{1}{r \sin \theta} \frac{\partial v_\theta}{\partial \phi} - \frac{v_\phi}{r} \cot \theta$$

$$\{\nabla \mathbf{v}\}_{,\phi\phi} = \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_r}{r} + \frac{v_\theta}{r} \cot \theta$$

$$\tau = -\mu(|\nabla \mathbf{v}| + |\nabla \mathbf{v}|^T) + \left(\frac{2}{3}\mu - \kappa\right)(\text{div } \mathbf{v})|\delta|$$

$$\nabla \mathbf{v} = \begin{bmatrix} v_{rr} & v_{r\theta} & v_{r\phi} \\ v_{\theta r} & v_{\theta\theta} & v_{\theta\phi} \\ v_{\phi r} & v_{\phi\theta} & v_{\phi\phi} \end{bmatrix} \quad \nabla \mathbf{v}^T = \begin{bmatrix} v_{rr} & v_{\theta r} & v_{\phi r} \\ v_{r\theta} & v_{\theta\theta} & v_{\phi\theta} \\ v_{r\phi} & v_{\theta\phi} & v_{\phi\phi} \end{bmatrix}$$

$$\tau_{r\phi} = -\mu \frac{\partial v_\phi}{\partial r} = -\mu (0) = 0$$

$$\tau_{r\theta} = -\mu \frac{\partial v_\theta}{\partial r} = -\mu \frac{\partial}{\partial r} \left(v_\infty \left[-1 + \frac{3}{4} \left(\frac{R}{r} \right) + \frac{1}{4} \left(\frac{R}{r} \right)^3 \right] \sin \theta \right)$$

$$= -\mu \left(v_\infty \left[-1 - \frac{3}{4} \left(\frac{R}{r^2} \right) - \frac{1}{4} \left(\frac{R}{r^2} \right)^3 \right] \sin \theta \right)$$

$$\tau_{r\phi} = -\mu \frac{\partial v_\phi}{\partial r} = -\mu \frac{\partial}{\partial r} \left(v_\infty \left[1 - \frac{3}{2} \left(\frac{R}{r} \right) + \frac{1}{2} \left(\frac{R}{r} \right)^3 \right] \cos \theta \right)$$

$$= -\mu \left(v_\infty \left[1 + \frac{3}{2} \left(\frac{R}{r^2} \right) - \frac{1}{2} \left(\frac{R}{r^2} \right)^3 \right] \cos \theta \right)$$